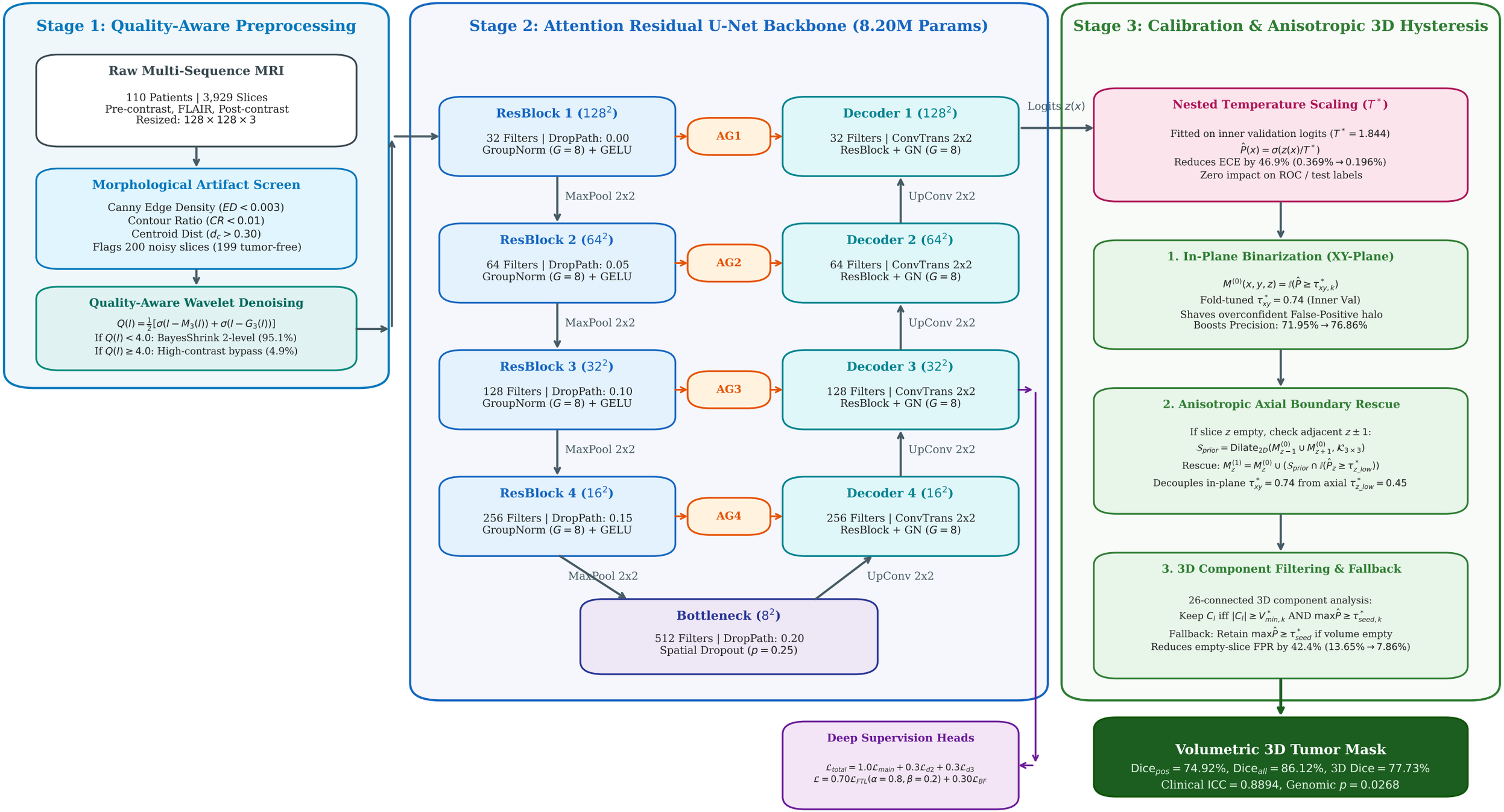

Stage 1: Quality-Aware Preprocessing
Raw Multi-Sequence MRI
110 Patients | 3,929 Slices
Pre-contrast, FLAIR, Post-contrast
Resized: 128 × 128 × 3
Morphological Artifact Screen
Canny Edge Density (ED < 0.003)
Contour Ratio (CR < 0.01)
Centroid Dist (d_c > 0.30)
Flags 200 noisy slices (199 tumor-free)
Quality-Aware Wavelet Denoising
Q(I) = ½[σ(I − M_3(I)) + σ(I − G_3(I))]
If Q(I) < 4.0: BayesShrink 2-level (95.1%)
If Q(I) ≥ 4.0: High-contrast bypass (4.9%)
Stage 2: Attention Residual U-Net Backbone (8.20M Params)
ResBlock 1 (128²)
32 Filters | DropPath: 0.00
GroupNorm (G = 8) + GELU
AG1
Decoder 1 (128²)
32 Filters | ConvTrans 2x2
ResBlock + GN (G = 8)
MaxPool 2x2
UpConv 2x2
ResBlock 2 (64²)
64 Filters | DropPath: 0.05
GroupNorm (G = 8) + GELU
AG2
Decoder 2 (64²)
64 Filters | ConvTrans 2x2
ResBlock + GN (G = 8)
MaxPool 2x2
UpConv 2x2
ResBlock 3 (32²)
128 Filters | DropPath: 0.10
GroupNorm (G = 8) + GELU
AG3
Decoder 3 (32²)
128 Filters | ConvTrans 2x2
ResBlock + GN (G = 8)
MaxPool 2x2
UpConv 2x2
ResBlock 4 (16²)
256 Filters | DropPath: 0.15
GroupNorm (G = 8) + GELU
AG4
Decoder 4 (16²)
256 Filters | ConvTrans 2x2
ResBlock + GN (G = 8)
MaxPool 2x2
UpConv 2x2
Bottleneck (8²)
512 Filters | DropPath: 0.20
Spatial Dropout (p = 0.25)
Logits z(x)
Deep Supervision Heads
L_total = 1.0L_main + 0.3L_d2 + 0.3L_d3
L = 0.70L_FTL(α = 0.8, β = 0.2) + 0.30L_BF
Stage 3: Calibration & Anisotropic 3D Hysteresis
Nested Temperature Scaling (T*)
Fitted on inner validation logits (T* = 1.844)
P̂(x) = σ(z(x)/T*)
Reduces ECE by 46.9% (0.369% → 0.196%)
Zero impact on ROC / test labels
1. In-Plane Binarization (XY-Plane)
M^(0)(x, y, z) = 𝕀(P̂ ≥ τ*_xy,k)
Fold-tuned τ*_xy = 0.74 (Inner Val)
Shaves overconfident False-Positive halo
Boosts Precision: 71.95% → 76.86%
2. Anisotropic Axial Boundary Rescue
If slice z empty, check adjacent z ± 1:
S_prior = Dilate_2D(M^(0)_z−1 ∪ M^(0)_z+1, K_3×3)
Rescue: M^(1)_z = M^(0)_z ∪ (S_prior ∩ 𝕀(P̂_z ≥ τ*_z_low))
Decouples in-plane τ*_xy = 0.74 from axial τ*_z_low = 0.45
3. 3D Component Filtering & Fallback
26-connected 3D component analysis:
Keep C_l iff |C_l| ≥ V*_min,k AND max P̂ ≥ τ*_seed,k
Fallback: Retain max P̂ ≥ τ*_seed if volume empty
Reduces empty-slice FPR by 42.4% (13.65% → 7.86%)
Volumetric 3D Tumor Mask
Dice_pos = 74.92%, Dice_all = 86.12%, 3D Dice = 77.73%
Clinical ICC = 0.8894, Genomic p = 0.0268